

Supplementary Online Content

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eTable 1. Numbers and Reasons for ineligibility

	Numbers
Total screenings done	9,232
Total eligible	1,086
Consent not given	6
Not willing to follow study procedures	3
Not willing to participate in study	3
Total ineligible	8,146
Age less than 15 years	385
Not diagnosed with pulmonary TB	2,783
Diagnosis not within last 4 weeks	1,820
Not a current smoker (not smoking at least 25 days in past one month)	7,069
Not willing to quit smoking	273
Not have access to mobile phone	1,290
Cannot read SMS message or and have no one at home to read the text message	236

eTable 2. Distribution of prior quit attempts by site and study arm

mHealth group			Usual care group		
Site ID	Yes, n (%)	No, n (%)	Site ID	Yes, n (%)	No, n (%)
1002	13 (32.5)	27 (67.5)	1001	10 (25.0)	30 (75.0)
1003	0 (0.0)	40 (100.0)	1004	1 (2.5)	39 (97.5)
1005	7 (17.5)	33 (82.5)	1009	0 (0.0)	40 (100.0)
1006	0 (0.0)	40 (100.0)	1010	0 (0.0)	40 (100.0)
1007	23 (57.5)	17 (42.5)	1014	5 (12.5)	35 (87.5)
1008	9 (22.5)	31 (77.5)	2002	3 (7.5)	37 (92.5)
1011	16 (40.0)	24 (60.0)	2004	0 (0.0)	40 (100.0)
1012	11 (27.5)	29 (72.5)	2008	4 (10.0)	36 (90.0)
1013	1 (2.5)	39 (97.5)	2011	10 (25.0)	30 (75.0)
1015	4 (10.0)	36 (90.0)			
2001	9 (22.5)	31 (77.5)			
2003	11 (27.5)	29 (72.5)			
2005	19 (47.5)	21 (52.5)			
2006	18 (45.0)	22 (55.0)			
2007	5 (12.5)	35 (87.5)			
2009	6 (15.0)	34 (85.0)			
2010	16 (40.0)	24 (60.0)			

eTable 3. Secondary outcome of TB treatment adherence

Month	m-Health		Control		z	P value
	Median(IQR)	Mean(SD)	Median(IQR)	Mean(SD)		
Month 1	30(30-30)	29.87(1.345)	30(30-30)	29.9(0.2)	-0.86	0.388
Month 2	30(30-30)	29.59(2.757)	30(30-30)	29.8(0.6)	0.44	0.656
Month 3	30(30-30)	29.39(3.894)	30(30-30)	29.7(2.3)	1.64	0.101
Month 4	30(30-30)	29(5.092)	30(30-30)	29.5(3.3)	0.44	0.657
Month 5	30(30-30)	28.56(6.049)	30(30-30)	29.5(3.3)	0.95	0.34
Month 6	30(30-30)	27.88(7.397)	30(30-30)	29.3(3.9)	1.85	0.064
Total	180(180-180)	174.3(21.501)	180(180-180)	178.0(12.1)	1.19	0.232

eTable 4. Causes of deaths among two study groups

Causes of death	n(%)		
	Overall	mHealth Group	Usual Care Group
Total deaths	52	25	27
Death due to TB	32(61.5%)	16(64.0%)	16(59.2%)
Cancer	3(5.7%)	0(0.0%)	3(11.1%)
Heart attack	8(15.2%)	5(20.0%)	3(11.1%)
History of fall, fever before death	2(3.8%)	1(4.0%)	1(3.7%)
Stroke	2(3.8%)	0(0.0%)	2(7.4%)
Liver failure	2(3.8%)	2(8.0%)	0(0.0%)
Drug user	1(1.9%)	0(0.0%)	1(7.4%)
HIV/AIDS comorbidity	1(1.9%)	1(4.0%)	0(0.0%)
Severe pneumonia	1(1.9%)	0(0.0%)	1(7.4%)

eFigure. Kaplan-Meier plot showing survival probability in two study groups

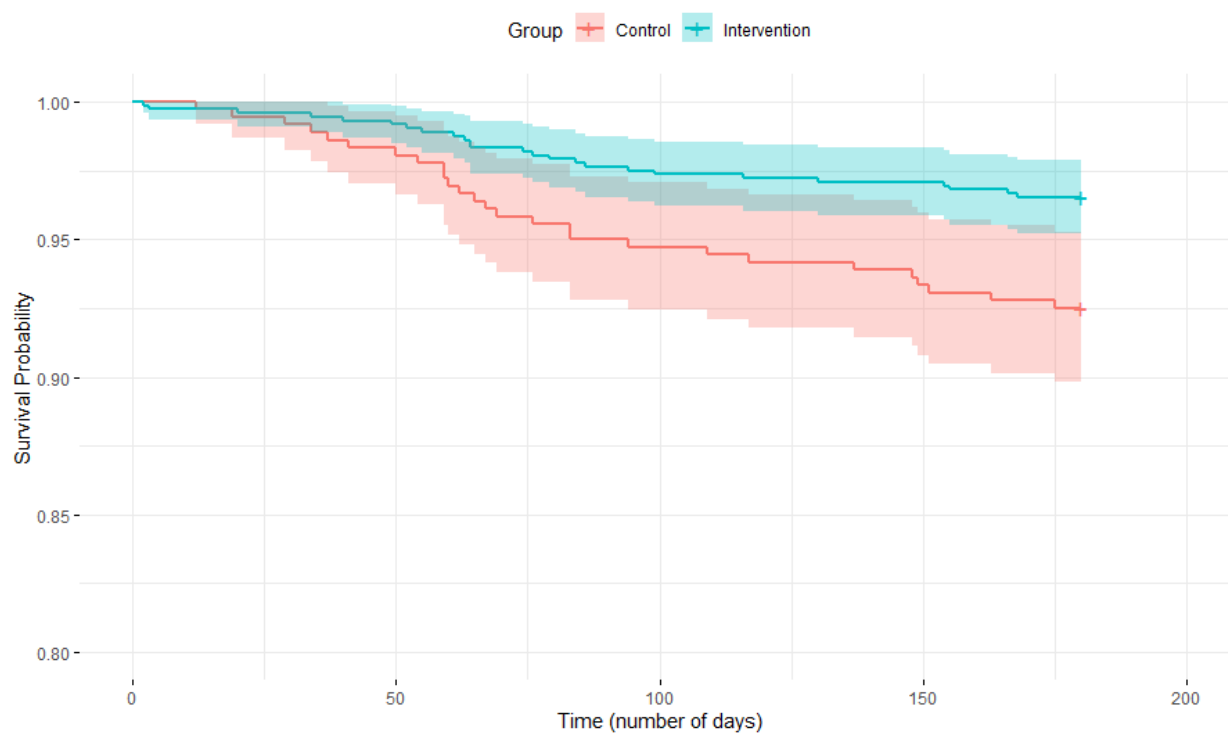


Figure S1: Kaplan-Meier plot showing survival probability in two study groups

eTable 5. Cluster specific recruitment and abstinence rates

	Site IDs	Recruitment Rate n/N (%)*	Self-reported Quitter at 6-mo FU n/N (%)	Verified Quitter at 6-mo FU n/N (%)	Quitter at 6-mo FU (ITT) n/N (%)
Control	1001	40/191(20.9%)	4/38(10.5)	4/38(10.5)	4/40(10)
m-Health	1002	40/250(16.0%)	17/40(42.5)	17/40(42.5)	17/40(42.5)
m-Health	1003	40/278(14.4%)	38/38(100)	35/38(92.1)	35/40(87.5)
Control	1004	40/226(17.7%)	7/36(19.4)	7/36(19.4)	7/40(17.5)
m-Health	1005	40/83(48.2%)	26/37(70.3)	25/37(67.6)	25/40(62.5)
m-Health	1006	40/111(36.0%)	6/40(15)	5/40(12.5)	5/40(12.5)
m-Health	1007	40/299(13.4%)	7/35(20)	7/35(20)	7/40(17.5)
m-Health	1008	40/202(19.8%)	10/39(25.6)	10/39(25.6)	10/40(25)
Control	1009	40/144(27.8%)	5/36(13.9)	5/36(13.9)	5/40(12.5)
Control	1010	40/211(19.0%)	13/37(35.1)	9/37(24.3)	9/40(22.5)
m-Health	1011	40/179(22.3%)	29/39(74.4)	29/39(74.4)	29/40(72.5)
m-Health	1012	40/271(14.8%)	23/39(59)	23/39(59)	23/40(57.5)
m-Health	1013	40/234(17.1%)	16/36(44.4)	16/36(44.4)	16/40(40)
Control	1014	40/170(23.5%)	7/38(18.4)	7/38(18.4)	7/40(17.5)
m-Health	1015	40/110(36.4%)	25/38(65.8)	22/38(57.9)	22/40(55)
m-Health	2001	40/1135(3.5%)	6/36(16.7)	5/36(13.9)	5/40(12.5)
Control	2002	40/706(5.6%)	9/33(27.3)	6/33(18.2)	6/40(15)
m-Health	2003	40/452(8.8%)	19/38(50)	16/38(42.1)	16/40(40)
Control	2004	40/379(10.5%)	3/35(8.6)	0/35(0)	0/40(0)
m-Health	2005	40/527(7.5%)	10/33(30.3)	6/33(18.2)	6/40(15)
m-Health	2006	40/563(7.1%)	17/36(47.2)	11/36(30.6)	11/40(27.5)
m-Health	2007	40/376(10.6%)	12/36(33.3)	12/36(33.3)	12/40(30)
Control	2008	40/484(8.2%)	15/35(42.9)	12/35(34.3)	12/40(30)
m-Health	2009	40/642(6.2%)	29/39(74.4)	26/39(66.7)	26/40(65)
m-Health	2010	40/424(9.4%)	25/34(73.5)	15/34(44.1)	15/40(37.5)
Control	2011	40/354(11.3%)	7/30(23.3)	5/30(16.7)	5/40(12.5)
m-Health	2012	40/239(16.7%)	27/34(79.4)	20/34(58.8)	20/40(50)

eTable 6. Cluster-wise death rates and unsuccessful TB treatment outcomes

	Site IDs	Deaths n(%)
Control	1001	1(2.5)
m-Health	1002	0(0.0)
m-Health	1003	0(0.0)
Control	1004	0(0.0)
m-Health	1005	2(5.0)
m-Health	1006	0(0.0)
m-Health	1007	0(0.0)
m-Health	1008	0(0.0)
Control	1009	4(10.0)
Control	1010	2(5.0)
m-Health	1011	1(2.5)
m-Health	1012	1(2.5)
m-Health	1013	4(10.0)
Control	1014	2(5.0)
m-Health	1015	0(0.0)
m-Health	2001	0(0.0)
Control	2002	3(7.5)
m-Health	2003	2(5.0)
Control	2004	4(10.0)
m-Health	2005	3(7.5)
m-Health	2006	1(2.5)
m-Health	2007	3(7.5)
Control	2008	5(7.5)
m-Health	2009	0(0.0)
m-Health	2010	3(7.5)
Control	2011	6(15.0)
m-Health	2012	5(12.5)

eTable 7. Cluster Characteristics and Associated Unadjusted Relative Risks

	Site ID	Age Mean(SD),y	Male, n(%)	Smoking Duration m(SD), y	Education			Occupation			
					No formal education	Primary	Middle	Higher	Employed	Dependent	Retired
Control	100 1	49.2(15.3)	40(100)	30.1(16.8)	16(40)	7(17.5)	7(17.5)	10(25)	37(92.5)	3(7.5)	0(0)
m-Health	100 2	40.3(14.9)	40(100)	19.5(12.2)	0(0)	16(40)	13(32.5)	11(27.5)	39(97.5)	1(2.5)	0(0)
m-Health	100 3	43.4(14.4)	40(100)	18(9.7)	7(17.5)	11(27.5)	13(32.5)	9(22.5)	36(90)	1(2.5)	3(7.5)
Control	100 4	37.9(12)	40(100)	20.6(12.7)	16(40)	14(35)	4(10)	6(15)	31(77.5)	7(17.5)	2(5)
m-Health	100 5	49.6(16.5)	40(100)	28.4(16.3)	17(42.5)	12(30)	7(17.5)	4(10)	34(85)	5(12.5)	1(2.5)
m-Health	100 6	57.5(14)	40(100)	37.1(14.5)	22(55)	6(15)	4(10)	8(20)	34(85)	0(0)	6(15)
m-Health	100 7	39.5(12.9)	40(100)	15.5(10.8)	11(27.5)	15(37.5)	7(17.5)	7(17.5)	33(82.5)	7(17.5)	0(0)
m-Health	100 8	54.2(12.5)	40(100)	34.4(11.9)	31(77.5)	5(12.5)	1(2.5)	3(7.5)	16(40)	24(60)	0(0)
Control	100 9	46.6(15)	40(100)	26.3(14.6)	0(0)	32(80)	3(7.5)	5(12.5)	30(75)	0(0)	10(25)
Control	101 0	49.6(17.8)	40(100)	30.2(16.8)	24(60)	6(15)	6(15)	4(10)	30(75)	9(22.5)	1(2.5)
m-Health	101 1	44.9(14.6)	40(100)	21.4(11.7)	12(30)	11(27.5)	6(15)	11(27.5)	29(72.5)	3(7.5)	8(20)
m-Health	101 2	37.4(14.8)	40(100)	17.7(14.3)	14(35)	10(25)	9(22.5)	7(17.5)	33(82.5)	4(10)	3(7.5)
m-Health	101 3	48.8(13.9)	40(100)	29(13.1)	18(45)	17(42.5)	1(2.5)	4(10)	37(92.5)	3(7.5)	0(0)
Control	101 4	55.9(11.6)	40(100)	33.4(11.3)	1(2.5)	19(47.5)	17(42.5)	3(7.5)	40(100)	0(0)	0(0)
m-Health	101 5	52(13.7)	39(97.5)	30.3(13.4)	25(62.5)	6(15)	4(10)	5(12.5)	37(92.5)	2(5)	1(2.5)

m-Health	200 1	39.3(13.9)	40(100)	16.4(14.1)	11(27.5)	8(20)	4(10)	17(42.5)	34(85)	6(15)	0(0)
Control	200 2	48.2(18.1)	40(100)	19(11.2)	17(42.5)	15(37.5)	1(2.5)	7(17.5)	28(70)	11(27.5)	1(2.5)
m-Health	200 3	53.5(18.1)	38(95)	29.8(18.3)	16(40)	12(30)	5(12.5)	7(17.5)	24(60)	16(40)	0(0)
Control	200 4	48.1(16.9)	34(85)	12.8(8.2)	30(75)	6(15)	1(2.5)	3(7.5)	28(70)	12(30)	0(0)
m-Health	200 5	50.2(15)	39(97.5)	23.6(15)	20(50)	5(12.5)	6(15)	9(22.5)	33(82.5)	1(2.5)	6(15)
m-Health	200 6	51.3(15.2)	38(95)	29.7(14.8)	31(77.5)	4(10)	2(5)	3(7.5)	31(77.5)	9(22.5)	0(0)
m-Health	200 7	56.5(15)	33(82.5)	28.7(14.3)	27(67.5)	5(12.5)	6(15)	2(5)	23(57.5)	17(42.5)	0(0)
Control	200 8	48.7(15.5)	32(80)	17.1(13.2)	21(52.5)	10(25)	1(2.5)	8(20)	22(55)	18(45)	0(0)
m-Health	200 9	48.2(16.3)	35(87.5)	18.9(14.8)	11(27.5)	13(32.5)	11(27.5)	5(12.5)	19(47.5)	18(45)	3(7.5)
m-Health	201 0	56.9(14.6)	36(90)	27.4(15.3)	23(57.5)	9(22.5)	7(17.5)	1(2.5)	38(95)	1(2.5)	1(2.5)
Control	201 1	54.7(14.6)	39(97.5)	33.2(16.7)	30(75)	5(12.5)	3(7.5)	2(5)	29(72.5)	11(27.5)	0(0)
m-Health	201 2	53.3(11.8)	40(100)	21.5(14.2)	19(47.5)	10(25)	3(7.5)	8(20)	30(75)	10(25)	0(0)
Relative risk (95% CI)		1.03 (1.01-1.04)	0.83 (0.34-2.02)	0.97 (0.95-0.99)	1	1.32 (0.88-1.97)	1.91 (1.16-3.15)	1.86 (1.16-2.99)	1	2.01 (1.30-3.09)	1.00 (0.43-2.31)

eTable 8. Subgroup analysis

Unadjusted Relative Risks among sub-groups for verified abstinence at month 6			
Variable	Category	RR	95CI%
All		2.890	(1.983 - 4.709)
Age	<40	2.672	1.472 - 4.857
	≥40	2.953	2.048 - 5.092
Education	No formal education	2.880	1.566 - 5.542
	Primary (year 1-5)	2.638	1.849 - 4.07
	Secondary or above (year ≥6)	2.719	1.348 - 4.83
Occupation	Active in job/business	2.989	1.933 - 4.885
	Dependent or retired	2.587	1.329 - 3.986
Smoking Duration	<24 Years	3.511	1.884 - 7.127
	≥24 Years	2.446	1.550 - 3.911
Reading SMS	Yes	2.769	1.743 - 4.318
	No	2.198	1.288 - 3.299

eTable 9. Post-hoc sensitivity analysis (intention to treat) after excluding deaths

Outcomes	n/N ^a		Ab diff. (95% CI) ^b	Crude RR ^c (95% CI)	Crude ICC ^d	Adj ^e RR (95% CI)	Adj ^e ICC
	% (95% CI)						
	mHealth	Usual Care					
Primary outcomes							
Biochemically verified abstinence at month-6, <10ppm ^f	300/695	55/333	26.7	2.9	0.18	3.1	0.18
	43.2(39.4-46.9)	16.5(12.7-20.9)	(21.2-32.1)	(1.8-6.4)		(1.9-6.5)	
Biochemically verified abstinence, <6ppm ^f	264/695	38/333	26.6	3.6	0.17	3.8	0.18
	38.0(34.4-41.7)	11.4(8.2-15.3)	(21.6-31.5)	(2.4-5.4)		(2.4-6.2)	
Secondary outcomes							
Point abstinence (last 7 days) at week-9	353/695	75/333	28.3	2.5	0.20	2.6	0.20
	50.8 (47.0- 54.6)	22.5 (18.1-27.4)	(22.4-34.1)	(1.7-3.6)		(1.7-3.8)	
Point abstinence last 7 days) at month-6	400/695	82/333	32.9	2.58	0.20	2.7	0.20
	57.5(53.8-61.2)	24.6(20.1-29.6)	(27.0-38.8)	(1.8-3.6)		(1.8-4.0)	
Successful TB Treatment (cured + completed treatment)	643/695	308/333	0	1.1	0.25	1.1	0.23
	92.5(90.3-94.4)	92.5(89.1-95.1)	(-3.4-3.5)	(0.7-1.6)		(0.7-1.5)	
Defaulted	22/695	7/333	1.1	-	-	-	-
	3.2(2.0-4.8)	2.1(0.8-4.3)	(-1.0-3.1)				
Treatment Failures	1/695	2/333	0.5	-	-	-	-
	0.1(0.01-0.8)	0.6(0.1-2.2)	(-0.4-1.3)				

^a Number of outcomes / total number in the group, ^b Absolute difference, ^c Relative risk, ^d Intraclass correlation coefficient, ^e Adjusted for age, sex, education, occupation, smoking duration: Adjusted analyses accounted for clustering and mixed-effects models for RR ^f Carbon-monoxide breath test cut-off values

eTable 10. Adverse Events

Symptoms		m-Health n(%)	Control n(%)	X2	P value
Nausea	None	538(77)	263(78.7)	6.5 ^a	0.084
	Mild	124(17.7)	64(19.2)		
	Moderate	31(4.4)	7(2.1)		
	Severe	6(0.9)	0(0)		
Diarrhoea	None	648(92.7)	309(92.5)	1.0 ^a	0.825
	Mild	43(6.2)	22(6.6)		
	Moderate	6(0.9)	3(0.9)		
	Severe	2(0.3)	0(0)		
Dry mouth	None	261(37.3)	148(44.3)	31.2	<.001
	Mild	267(38.2)	154(46.1)		
	Moderate	116(16.6)	23(6.9)		
	Severe	55(7.9)	9(2.7)		
Epigastric pain	None	505(72.3)	199(59.6)	18.2 ^a	<.001
	Mild	143(20.5)	107(32)		
	Moderate	43(6.2)	24(7.2)		
	Severe	7(1)	4(1.2)		
Headache	None	384(54.9)	170(50.9)	2.7	0.426
	Mild	227(32.5)	126(37.7)		
	Moderate	70(10)	30(9)		
	Severe	18(2.6)	8(2.4)		
Insomnia	None	452(64.7)	222(66.5)	6.9	0.072
	Mild	170(24.3)	87(26)		
	Moderate	55(7.9)	23(6.9)		
	Severe	22(3.1)	2(0.6)		
Abnormal dreams	None	629(90)	290(86.8)	3.8 ^a	0.255
	Mild	55(7.9)	38(11.4)		
	Moderate	14(2)	6(1.8)		
	Severe	1(0.1)	0(0)		
Irritability	None	416(59.5)	189(56.6)	18.5	<.001
	Mild	175(25)	117(35)		
	Moderate	78(11.2)	24(7.2)		
	Severe	30(4.3)	4(1.2)		
Anxiety	None	466(66.7)	211(63.2)	17.1	<.001
	Mild	168(24)	110(32.9)		
	Moderate	48(6.9)	12(3.6)		
	Severe	17(2.4)	1(0.3)		
Palpitations	None	482(69)	239(71.6)	5.2 ^a	0.154
	Mild	168(24)	73(21.9)		

	Moderate	40(5.7)	22(6.6)		
	Severe	9(1.3)	0(0)		
Musculoskeletal pain	None	270(38.6)	131(39.2)	8.8	0.031
	Mild	251(35.9)	143(42.8)		
	Moderate	134(19.2)	48(14.4)		
	Severe	44(6.3)	12(3.6)		

a: based on Exact test